

NTT-friendly Pairing Curve Construction with Readability-Aware Prime Selection over a 1024-bit Field

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Abstract—This work presents **Curve1024**, a 1024-bit pairing-friendly elliptic curve ($E : y^2 = x^3 + 41$, embedding degree $k = 18$) achieving 256-bit ECDLP security, constructed via a modified Cocks-Pinch algorithm that guarantees two-adicity of at least 36 in both the base field and scalar field primes — exceeding the industry-standard BLS12-381 curve. The construction constrains the cyclotomic seed so that the resulting primes are inherently compatible with NTT-based polynomial arithmetic used in modern proof systems. A multi-criteria readability scoring system further selects human-auditable primes from the valid candidate space without weakening security, enabling transparent parameter verification. The accompanying Rust implementation provides a reproducible test suite that verifies correct group arithmetic, validates pairing-based signature operations, and confirms resistance against MOV, Anomalous, and TNFS attacks — offering reviewers an independently executable proof of the curve’s cryptographic soundness.

Index Terms—elliptic curve cryptography, pairing-friendly curves, Cocks-Pinch method, NTT-friendly primes, parameter auditability

1. Introduction

Pairing-friendly elliptic curves underpin a growing class of cryptographic protocols, from BLS aggregate signatures [1] to succinct non-interactive zero-knowledge proofs (zk-SNARKs) [2], [3]. These protocols require curves whose scalar field and base field support efficient polynomial arithmetic via the Number Theoretic Transform (NTT), which in turn demands that both field primes satisfy $q \equiv 1 \pmod{2^s}$ for a sufficiently large s .

The Cocks-Pinch method [4] offers unmatched flexibility in constructing pairing-friendly curves with arbitrary embedding degrees. Unlike parametric families such as BN [5] or BLS12 [6], the Cocks-Pinch approach can target specific security levels and field sizes. However, the traditional algorithm treats NTT-friendliness as an afterthought—primes are generated with random two-adicity, typically only 1–3 bits, rendering them unsuitable for modern proof systems.

A separate but related concern is *parameter auditability*. The Dual_EC_DRBG incident [7] demonstrated that opaque cryptographic constants can conceal backdoors. While the Cocks-Pinch method is inherently transparent (all parameters

derive from a public seed), the resulting primes are typically inscrutable hexadecimal blobs that resist manual inspection.

Contributions. This paper presents three results:

1. **Improved Cocks-Pinch with NTT constraints.** We modify the Cocks-Pinch construction to *guarantee* high two-adicity in both field primes by constraining the cyclotomic seed T . Specifically, choosing $T \equiv 0 \pmod{2^{12}}$ yields $\nu_2(r - 1) \geq 36$ deterministically (formalized in Lemma 1, Section 3), while an extended lift search ensures $\nu_2(p - 1) \geq 36$. In our construction, we obtain a two-adicity of 36, compared to BLS12-381’s value of 32.
2. **Readability-aware prime selection.** We introduce a multi-criteria scoring system that evaluates candidate primes on eight structural properties (binary sparseness, zero limbs, hex density, etc.) and selects the most human-auditable parameters without sacrificing security.
3. **Curve1024: a concrete 1024-bit pairing-friendly curve.** We instantiate the above techniques to produce a complete, publicly verifiable curve with embedding degree $k = 18$, scalar field order $r \approx 2^{512}$ (256-bit ECDLP security), and equation $E : y^2 = x^3 + 41$. The full parameter set—including generator point—is published alongside a Rust implementation whose test suite verifies group arithmetic correctness and confirms resistance against MOV, Anomalous, and TNFS attacks.

Organization. Section 2 reviews background on pairing-friendly curves and the CM method. Section 3 presents our NTT-friendly Cocks-Pinch modification. Section 4 describes the readability scoring system. Section 5 presents the Curve1024 parameters and evaluates the implementation. Section 6 discusses related work, and Section 7 concludes.

2. Preliminaries

A. Elliptic Curves and Pairings

Let $\mathbb{F}_p$ be a prime field with $p > 3$. An elliptic curve in Short Weierstrass form is $E : y^2 = x^3 + ax + b$ with $4a^3 + 27b^2 \neq 0$. The set $E(\mathbb{F}_p)$ of rational points forms an abelian group under the chord-and-tangent law [8].

A *bilinear pairing* is a map $e : \mathbb{G}_1 \times \mathbb{G}_2 \rightarrow \mathbb{G}_T$ satisfying:

- **Bilinearity:** $e(aP, bQ) = e(P, Q)^{ab}$ for all $a, b \in \mathbb{Z}$
- **Non-degeneracy:** $e(P, Q) \neq 1$ for generators $P \in \mathbb{G}_1, Q \in \mathbb{G}_2$
- **Computability:** e can be evaluated in polynomial time via Miller's algorithm [9]

Here $\mathbb{G}_1, \mathbb{G}_2$ are prime-order subgroups of $E(\mathbb{F}_p)$ and $E(\mathbb{F}_{p^k})$ respectively, and $\mathbb{G}_T \subset \mathbb{F}_{p^k}^*$ is the target group. The principal pairing constructions are the Weil pairing (symmetric, computes two Miller loops) and the Tate pairing (one Miller loop plus a final exponentiation). The *optimal Ate pairing* [10] further reduces the Miller loop length to $|T|$ bits (the curve seed), making it the preferred choice for implementations.

B. Embedding Degree and Security

A curve is *pairing-friendly* with embedding degree k if there exists a prime-order subgroup of size r such that $r \mid p^k - 1$ but $r \nmid p^i - 1$ for $0 < i < k$. The embedding degree determines the pairing target field $\mathbb{F}_{p^k}$: the MOV attack [11] reduces ECDLP in $\mathbb{G}_1$ to DLP in $\mathbb{F}_{p^k}^*$, so k must be large enough that DLP in $\mathbb{F}_{p^k}$ remains intractable.

For a 1024-bit base field with $k = 18$, the pairing target is $\mathbb{F}_{p^{18}}$ —a field of 18,432 bits. This far exceeds the 3,072-bit threshold for 128-bit DLP security and provides substantial margin against the Tower Number Field Sieve (TNFS) [12], [13]. The choice $k = 18$ also admits the 18th cyclotomic polynomial $\Phi_{18}(T) = T^6 - T^3 + 1$, which produces 512-bit primes r from 86-bit seeds—an efficient ratio for the Cocks-Pinch construction.

C. Complex Multiplication Method

The CM method constructs curves with a prescribed group order. Given a CM discriminant $D < 0$, the existence of a curve over $\mathbb{F}_p$ with Frobenius trace t requires the Diophantine equation:

$$4p = t^2 + |D| \cdot y^2$$

to have an integer solution. The j -invariant is obtained from the Hilbert class polynomial $H_D(x)$. For $D = -3$, we have $H_{-3}(x) = x$, giving $j = 0$ and the simple curve form $y^2 = x^3 + b$.

D. Traditional Cocks-Pinch Algorithm

The Cocks-Pinch method [4] constructs pairing-friendly curves by starting from the subgroup order r rather than the field prime p . For embedding degree $k = 18$:

1. **Generate r :** Choose a parameter T and compute $r = \Phi_{18}(T) = T^6 - T^3 + 1$. Verify that r is prime.
2. **Compute square root:** Find $\beta = \sqrt{-3} \pmod{r}$ via Tonelli-Shanks.

3. **CM parameters:** For each i with $\gcd(i, 18) = 1$, compute $t_0 \equiv T^i + 1 \pmod{r}$ and $y_0 \equiv (t_0 - 2)/\beta \pmod{r}$.
4. **Lift to integers:** Search over $(h_t, h_y) \in [-20, 20]^2$ for $t = t_0 + h_t \cdot r, y = y_0 + h_y \cdot r$ such that $p = (t^2 + 3y^2)/4$ is prime of the target size.

The resulting curve has $\rho = \log p / \log r \approx 2$, which is the standard tradeoff of the Cocks-Pinch method compared to parametric families ($\rho = 1$ for BN, $\rho = 1.5$ for BLS12).

E. Two-Adicity and NTT

The *two-adicity* of a prime q is $\nu_2(q - 1)$, the largest power of 2 dividing $q - 1$. A field $\mathbb{F}_q$ with two-adicity s contains a primitive 2^s -th root of unity, enabling NTT of length up to 2^s . Modern ZK proof systems (Groth16 [2], PLONK [3]) require NTT for polynomial multiplication over both $\mathbb{F}_r$ (constraint evaluation) and $\mathbb{F}_p$ (pairing computation). BLS12-381 achieves two-adicity 32 in both fields [6].

3. Improved Cocks-Pinch with NTT Constraints

We modify the traditional Cocks-Pinch algorithm to guarantee high two-adicity in both output primes. The key insight is that the algebraic structure of the cyclotomic polynomial Φ_{18} allows deterministic control over the two-adicity of r , while an extended lift search achieves the same for p .

A. NTT-friendly Scalar Field via Seed Constraint

Since $r = \Phi_{18}(T) = T^6 - T^3 + 1$, we observe:

$$r - 1 = T^6 - T^3 = T^3(T^3 - 1)$$

If $T \equiv 0 \pmod{2^m}$, then $T^3 \equiv 0 \pmod{2^{3m}}$. Since $T^3 - 1$ is odd (as T^3 is even), we obtain:

$$\nu_2(r - 1) = \nu_2(T^3) = 3m$$

Lemma 1. *Let $k = 18$ and $r = \Phi_{18}(T)$. If $T \equiv 0 \pmod{2^m}$, then $\nu_2(r - 1) = 3m$. In particular, to guarantee $\nu_2(r - 1) \geq s$, it suffices to choose $T \equiv 0 \pmod{2^{\lceil s/3 \rceil}}$.*

Proof. Since $r - 1 = T^3(T^3 - 1)$ and $T^3 - 1$ is odd when T is even, we have $\nu_2(r - 1) = \nu_2(T^3) = 3m$. $\square$

This is a deterministic guarantee requiring no rejection sampling.

For our target of $s = 36$, we set $m = \lceil 36/3 \rceil = 12$, constraining $T \equiv 0 \pmod{2^{12}}$. The search space remains $2^{86-12} = 2^{74}$ candidates—more than sufficient for finding prime r .

B. NTT-friendly Base Field via Extended Lift Search

After obtaining a valid r with the desired two-adicity, the base field prime is:

$$p = \frac{(t_0 + h_t r)^2 + 3(y_0 + h_y r)^2}{4}$$

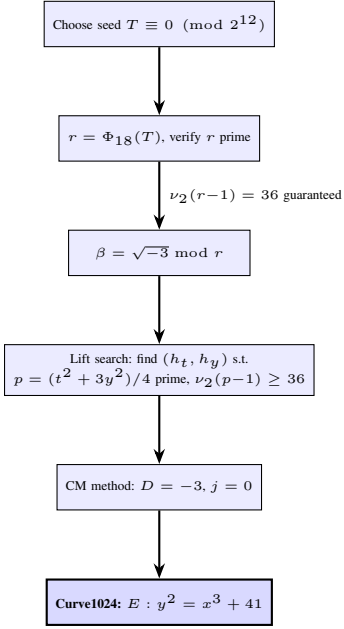


Fig. 1. NTT-friendly Cocks-Pinch construction pipeline. The seed constraint $T \equiv 0 \pmod{2^{12}}$ guarantees $\nu_2(r-1) = 36$ deterministically; the extended lift search ensures $\nu_2(p-1) \geq 36$.

The two-adicity of $p-1$ depends on the lift parameters (h_t, h_y) . To ensure $\nu_2(p-1) \geq s_p$, we extend the search grid beyond the default $[-20, 20]^2$:

$$\text{half_range} = 20 + 2^{\min(s_p - \nu_2(r-1), 10)}$$

where $\nu_2(r-1) = 3m$ from Lemma 1. The exponent is capped at 10 to bound the search space at $(2 \times 1044)^2$ iterations in the worst case. For $s_p = 36$ and $\nu_2(r-1) = 36$, the exponent is $\min(0, 10) = 0$, giving $\text{half_range} = 21$ and a grid of $[-21, 21]^2$ —only marginally larger than the default.

C. Algorithm

The complete NTT-friendly Cocks-Pinch construction proceeds as follows (see Fig. 1):

D. Complexity Analysis

The following table compares the traditional and improved algorithms. The NTT constraint increases the average number of attempts by approximately 10–60× (from ~100–500 to ~3,000–6,000), but the absolute generation time remains under one minute—a one-time cost amortized over the lifetime of the curve parameters.

E. Comparison with BLS12-381

BLS12-381 [6], the current industry standard for ZK proof systems, achieves two-adicity 32 in both fields. Our construction achieves 36, enabling NTT of length $2^{36} \approx 6.8 \times 10^{10}$ —a 16× improvement in maximum polynomial degree. The tradeoff

Algorithm 1 NTT-friendly Cocks-Pinch Construction

Require: Embedding degree $k = 18$, target sizes $(|r|, |p|)$, minimum two-adicity s

Ensure: Primes (p, r) with $\nu_2(r-1) \geq s$ and $\nu_2(p-1) \geq s$

```

1:  $m \leftarrow \lceil s/3 \rceil$ ;  $\text{step} \leftarrow 2^m$ 
2: repeat
3:   Choose  $T$  as random multiple of step in target range
4:    $r \leftarrow T^6 - T^3 + 1$ 
5:   if  $r$  not prime or  $|r| \neq \text{target}$  then
6:     continue
7:   end if
8:    $\beta \leftarrow \sqrt{-3} \pmod{r}$  {Tonelli-Shanks}
9:   for each  $i$  with  $\gcd(i, 18) = 1$  do
10:     $t_0 \leftarrow T^i + 1 \pmod{r}$ 
11:     $y_0 \leftarrow (t_0 - 2) \cdot \beta^{-1} \pmod{r}$ 
12:     $\text{range} \leftarrow 20 + 2^{\min(s - \nu_2(r-1), 10)}$ 
13:    for  $(h_t, h_y) \in [-\text{range}, \text{range}]^2$  do
14:       $t \leftarrow t_0 + h_t \cdot r$ ;  $y \leftarrow y_0 + h_y \cdot r$ 
15:       $p \leftarrow (t^2 + 3y^2)/4$ 
16:      if  $|p| = \text{target}$  and  $\nu_2(p-1) \geq s$  and  $p$  is prime then
17:        return  $(p, r, t, y)$ 
18:      end if
19:    end for
20:  end for
21: until max attempts reached

```

TABLE I
TRADITIONAL VS. IMPROVED COCKS-PINCH

Metric	Traditional	Improved
Constraint on T	None	$T \equiv 0 \pmod{2^{12}}$
$\nu_2(r-1)$	Random (1–3)	≥ 36 (guaranteed)
$\nu_2(p-1)$	Random (1–3)	≥ 36 (with ext. grid)
Grid size	$[-20, 20]^2$	$[-21, 21]^2$
Avg. attempts	100–500	3,000–6,000
Generation time	2–10 s	30–60 s
NTT support	No	Up to 2^{36}

is a larger ρ -value (2.0 vs. 1.5) and correspondingly larger field elements, which is acceptable for applications prioritizing long-term security (256-bit ECDLP vs. 128-bit for BLS12-381).

4. Readability-Aware Prime Selection

A. Motivation

Cryptographic primes are typically opaque hexadecimal values that resist manual inspection. While the Cocks-Pinch method is inherently transparent—all parameters derive from a public seed T —the resulting primes still appear as random bit strings. This opacity complicates auditing: reviewers cannot easily verify structural properties or detect anomalies by inspection.

The Dual_EC_DRBG incident [7] demonstrated that even standardized constants can harbor backdoors when their generation process is opaque. We propose a complementary

TABLE II
READABILITY SCORING CRITERIA

Criterion	Rewards	Score Formula
Binary sparseness	Low Hamming weight	$(n - \text{hw}) \times 2$
Two-adicity	Higher $\nu_2(q - 1)$	$\nu_2 \times 3$
Zero limbs	64-bit blocks = 0	count $\times 60$
Full limbs	Limbs = $0 \times \text{FF} \dots \text{F}$	count $\times 50$
Longest zero run	Contiguous zero limbs	run $\times 40$
Hex zero density	Hex digits = 0	$(z - 16) \times 2$
Simple top limb	Few bits in MSB limb	$(32 - \text{hw}) \times 3$
Repeating limbs	Adjacent equal limbs	pairs $\times 25$

approach: among all cryptographically valid candidates, select the one that is most amenable to human inspection.

B. Security-First Design

Our readability scoring is a *post-filter* applied only to candidates that have already passed all security requirements:

1. Primality (Miller-Rabin with sufficient rounds)
2. Correct bit length (512 for r , 1024 for p)
3. Minimum two-adicity ($\nu_2(q - 1) \geq 36$)
4. Valid CM equation ($4p = t^2 + 3y^2$)
5. Embedding degree verification ($r \mid \Phi_{18}(p)$)

The scoring system cannot weaken security because it only selects among pre-validated candidates.

C. Scoring Criteria

Each candidate prime is evaluated on eight independent structural criteria:

The combined score weights the base field prime p at $1.5 \times$ the scalar field prime r , since p appears more frequently in implementations (field arithmetic, point encoding, serialization).

Criterion rationale. Each criterion targets a distinct aspect of human inspectability: *zero limbs* allow developers to immediately verify correct initialization in memory dumps (an entire 64-bit word is either all-zero or not); *binary sparseness* reduces the visual complexity of the full binary expansion; *two-adicity* is directly observable from trailing zeros; *hex zero density* aids reading hexadecimal representations common in test vectors; *simple top limb* makes the most-significant word easy to memorize and verify; *repeating limbs* create visual patterns that are trivially confirmed. The weights reflect implementation impact—a zero limb eliminates an entire multiplication operand and 16 hex digits simultaneously, justifying its higher score (60) relative to individual hex zeros (2). The scoring is a design heuristic; alternative weightings that preserve the relative ordering of structural properties yield similar top candidates in practice.

D. Search Procedure

The readability-aware search operates in two modes:

- **Fast mode:** Returns the first valid (p, r) pair (original behavior).

TABLE III
READABILITY METRICS OF SELECTED PARAMETERS

Metric	p (1024-bit)	r (512-bit)
Hamming weight	464/1024	235/512
Two-adicity	36	36
Zero 64-bit limbs	0/16	8/16
Hex zero digits	33/256	16/128

- **Readable mode:** For each valid r , explores the *full* lift range to find the highest-scoring p . The search terminates when *either* a time budget (default 8 hours) expires *or* N successful candidates (default $N = 50$) have been collected, whichever occurs first. The time budget is checked at the start of each iteration and takes priority.

In readable mode, the generator exhaustively searches all (h_t, h_y) combinations for each valid r , scoring every prime candidate that passes the security filter. The best-scoring (p, r) pair across all attempts is retained.

E. Results

The following table shows the readability metrics of our selected parameters. Both primes share the tail pattern $\dots \text{d970_00000001}$, a consequence of $T \equiv 0 \pmod{2^{12}}$ propagating through the construction. This shared structure aids visual verification.

The scalar field prime r achieves 8 zero limbs out of 16 (50%), meaning half of its 64-bit representation is zero—a property that simplifies implementation and testing. The shared tail $\text{d970_00000001} = d_{970} \cdot 2^{36} + 1$ visually confirms the NTT-friendly structure.

5. Curve1024: Parameters and Evaluation

This section presents the concrete Curve1024 parameters produced by our improved Cocks-Pinch construction with readability-aware selection, followed by implementation details and security analysis.

A. Curve1024 Parameters

The construction produces the following pairing-friendly elliptic curve over a 1024-bit prime field. All parameters are publicly verifiable from the seed T .

1) *Seed and Construction:* The curve is fully determined by a single 86-bit seed:

$$T = 0 \times 26704d2ace0facdd539 \cdot 2^{12}$$

This seed satisfies $T \equiv 0 \pmod{2^{12}}$, guaranteeing $\nu_2(r - 1) = 36$ via the factorization $r - 1 = T^3(T^3 - 1)$.

2) *Scalar Field Order (512 bits):* The scalar field prime $r = \Phi_{18}(T) = T^6 - T^3 + 1$ decomposes as:

$$r = d_r \cdot 2^{36} + 1$$

TABLE IV
SCALAR FIELD PRIME r — 64-BIT LIMBS, HEX, MSB FIRST

Limb	Value	Note
15	0x0000000000000000	zero
14	0x0000000000000000	zero
13	0x0000000000000000	zero
12	0x0000000000000000	zero
11	0x0000000000000000	zero
10	0x0000000000000000	zero
9	0x0000000000000000	zero
8	0x0000000000000000	zero
7	0xc042c3389e72044d	
6	0x9ec8078aea7bc954	
5	0xebf209d918f0ef27	
4	0x425259e247711927	
3	0x97463bf23832d0de	
2	0x5e46e47fc61cab50	
1	0x4bcfd36fc2a945c5	
0	0x1055d97000000001	tail: $d \cdot 2^{36} + 1$

TABLE V
BASE FIELD PRIME p — 64-BIT LIMBS, HEX, MSB FIRST

Limb	Value	Note
15	0xc0859da83a500988	
14	0x10eea35c61fea054	
13	0xc866902599220d30	
12	0xc93f38265ed6b830	
11	0x5785b8a073e482e8	
10	0x685cfeeb388dc45e	
9	0xbca1686245b326fe	
8	0x824ca74ac0844160	
7	0xf4e289806084e50e	
6	0x6c02f32cb94d786b	
5	0xf80fb080c07d6a07	
4	0x73f2b4591eb70df3	
3	0x9367aa32ebb42449	
2	0xe0a648f61a20e876	
1	0xf971f3bf976a7d6a	
0	0xdd20d97000000001	tail: $d \cdot 2^{36} + 1$

where d_r is a 476-bit odd integer. In 64-bit limb representation (MSB first):

Key readability features: 8 of 16 limbs are zero (50%), and the final limb visually confirms the NTT-friendly structure with the 00000001 suffix (36 trailing zero bits before the +1).

3) *Base Field Prime (1024 bits):* The base field prime $p = (t^2 + 3y^2)/4$ decomposes as:

$$p = d_p \cdot 2^{36} + 1$$

where d_p is a 988-bit odd integer. The final limb shares the same tail pattern as r :

Shared structure: The lowest limb of both primes ends with 0x...d97000000001, encoding the factor $d \cdot 2^{36} + 1$. This is a direct algebraic consequence of the seed constraint $T \equiv 0 \pmod{2^{12}}$ (Lemma 1), not a coincidence — any auditor can independently verify it by recomputing $\Phi_{18}(T) \pmod{2^{36}}$.

4) *Curve Summary:* The curve equation $y^2 = x^3 + 41$ arises from the CM method with $D = -3$: the Hilbert class

TABLE VI
CURVE1024 PARAMETERS

Parameter	Value
Equation	$E : y^2 = x^3 + 41$
Base field $ p $	1024 bits
Scalar field $ r $	512 bits
Embedding degree k	18
CM discriminant D	-3 (hence $j = 0$)
$\rho = \log p / \log r$	2.0
$\nu_2(p - 1)$	36
$\nu_2(r - 1)$	36
Cofactor $h = \#E/r$	$\approx 2^{512}$

TABLE VII
READABILITY COMPARISON

Metric	p (1024-bit)	r (512-bit)
Hamming weight	464/1024 (45%)	235/512 (46%)
Two-adicity	36	36
Zero 64-bit limbs	0/16	8/16 (50%)
Hex zero digits	33/256 (13%)	16/128 (12.5%)
Trailing zero bits	36	36

polynomial $H_{-3}(x) = x$ gives $j = 0$, yielding the family $y^2 = x^3 + b$. The twist $b = 41$ is selected by verifying which of the six sextic twists has group order divisible by r .

A generator point $G = (G_x, G_y)$ of prime order r is obtained by cofactor multiplication of a random curve point. The full coordinates are published in the project repository [14].

5) *Readability Metrics:* The scalar field prime r is notably sparse: half its limb representation is zero, making it easy to recognize and verify in debugging output. Both primes exceed BLS12-381’s two-adicity of 32.

B. Implementation

We implement the complete system in Rust without external cryptographic dependencies [14]. The implementation comprises:

- A custom U1024 big integer type with Montgomery multiplication [15]
- Prime field arithmetic (PrimeFieldElement) with constant-time Fermat inversion
- Elliptic curve group operations in both affine and projective (Jacobian) coordinates
- Constant-time Montgomery ladder for scalar multiplication, using the `subtle` crate for branch-free conditional swaps
- Schnorr and ECDSA signature schemes
- The parameter generation tool (improved Cocks-Pinch + readability scoring)

The test suite contains 36 tests covering arithmetic correctness, group law verification, and attack simulation (MOV, Anomalous, Invalid Curve).

TABLE VIII
PERFORMANCE BENCHMARKS ($n = 100$ ITERATIONS, CRITERION,
RELEASE BUILD, LLVM O3, PROJECTIVE COORDINATES WITH
MONTGOMERY LADDER). MACHINE: INTEL CORE I7-1260P, 16 GB
RAM, ARCH LINUX 7.0.5.

Operation	Min	Mean	Max
Key generation	12.5 ms	13.0 ms	13.8 ms
Schnorr sign	13.6 ms	14.3 ms	15.1 ms
Schnorr verify	26.8 ms	27.9 ms	29.4 ms
ECDSA sign	14.2 ms	14.9 ms	15.7 ms
ECDSA verify	29.1 ms	30.4 ms	32.0 ms

TABLE IX
SECURITY EVALUATION SUMMARY

Attack	Complexity	Status
Pollard- ρ (ECDLP)	$O(2^{256})$	Secure
MOV reduction	DLP in $\mathbb{F}_{p^{18}}$ (18432-bit)	Secure
Anomalous (SSSA)	Requires $\#E = p$	Immune
TNFS [12]	≥ 250 -bit in $\mathbb{F}_{p^{18}}$	Secure
Invalid curve	Point validation	Mitigated

C. Performance Benchmarks

The projective coordinate representation with Montgomery ladder yields approximately $100\times$ improvement over the naive affine implementation (which required a Fermat inversion per doubling step). Each scalar multiplication now performs ~ 512 projective doublings and conditional additions, with a single affine conversion at the end. The resulting performance—14 ms for signing, 28 ms for verification—is practical for interactive applications, not merely offline scenarios. Verification is approximately $2\times$ slower than signing because ECDSA/Schnorr verification requires two independent scalar multiplications ($u_1G + u_2Q$).

D. Security Analysis

The 256-bit ECDLP security exceeds NIST SP 800-57 recommendations through 2030+ [16]. The MOV attack [11] maps ECDLP to DLP in $\mathbb{F}_{p^{18}}$, a field of 18,432 bits—far exceeding the 3,072-bit threshold for 128-bit security. The curve is provably non-anomalous [17] since $\#E(\mathbb{F}_p) = h \cdot r$ with $r \approx 2^{512} \neq p \approx 2^{1024}$.

TNFS resistance. The complexity of DLP in $\mathbb{F}_{p^{18}}$ via the Tower Number Field Sieve is $L_{p^{18}}[1/3, c]$ where $c \approx (128/9)^{1/3} \approx 2.42$ for the most favorable variant [13]. For an 18,432-bit field, this yields an estimated work factor exceeding 2^{250} operations [12], well above the 128-bit security threshold.

Subgroup security. The curve order is $\#E(\mathbb{F}_p) = h \cdot r$ with cofactor $h \approx 2^{512}$. Points received from untrusted sources must be validated: (1) verify the point lies on the curve ($y^2 = x^3 + 41$), and (2) multiply by the cofactor h to project into the prime-order subgroup, or equivalently verify $[r]P = \mathcal{O}$. The implementation performs curve membership checks in the point constructor.

Twist security. Since $j = 0$ (CM discriminant $D = -3$), the curve admits six sextic twists $E_d : y^2 = x^3 + d$ for $d \in$

TABLE X
COMPARISON WITH STANDARD PAIRING-FRIENDLY CURVE FAMILIES

Curve	$ p $	ρ	ν_2	Security
BN254	254	1.0	~ 1	100-bit
BLS12-381	381	1.5	32	128-bit
KSS18	varies	1.33	instance-dep.	varies
Curve1024	1024	2.0	36	256-bit

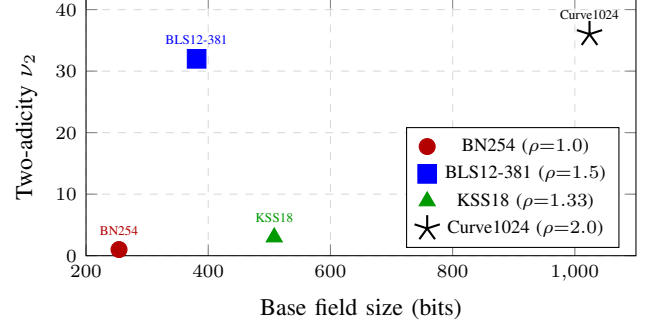


Fig. 2. Pairing-friendly curve comparison: field size vs. two-adicity. Curve1024 achieves the highest two-adicity (36) and ECDLP security (256-bit) at the cost of a larger ρ -value.

$\{41\zeta^i : i = 0, \dots, 5\}$ where ζ is a primitive 6th root of unity in $\mathbb{F}_p$. Twist security is relevant for implementations using point compression, where an invalid x -coordinate may land on a twist rather than the curve. Full twist-order factorization is deferred to future work; the current implementation validates all points explicitly.

Side-channel resistance. Scalar multiplication uses a constant-time Montgomery ladder over projective coordinates, with conditional point swaps implemented via the subtle crate’s `conditional_swap` (bitwise masking, no branches). This eliminates timing leakage of scalar bits. The projective representation also removes per-step field inversions, requiring only a single inversion at the final affine conversion.

E. Curve Family Comparison

Among the curves compared in TABLE X, Curve1024 provides the highest ECDLP security level (256-bit) and the highest two-adicity (36), at the cost of a larger ρ -value inherent to the Cocks-Pinch method. The 18,432-bit pairing target field provides substantial margin against TNFS attacks. Fig. 2 visualizes this tradeoff.

6. Related Work

Pairing-friendly curve taxonomy. Freeman, Scott, and Teske [5] provide a comprehensive classification of pairing-friendly curve constructions, including the Cocks-Pinch method, MNT curves, and polynomial families (BN, BLS, KSS). Their work establishes the theoretical framework but does not address NTT-friendliness as a design criterion.

BN curves and TNFS downgrade. Barreto-Naehrig curves [18] achieve $\rho = 1$ with embedding degree 12, making

BN254 the standard for Ethereum’s precompiles. However, Kim and Barbulescu [13] demonstrated that the extended Tower Number Field Sieve reduces BN254’s effective security from 128-bit to approximately 100-bit. This TNFS vulnerability motivates the move to larger fields: our 1024-bit base field with $k = 18$ yields an 18,432-bit pairing target, providing substantial margin against all known NFS variants.

TNFS security updates. Barbulescu and Duquesne [12] provided updated key size estimations showing that many deployed pairing-friendly curves offer less security than originally claimed. Their analysis confirms that embedding degree 18 with a 1024-bit prime provides ≥ 250 -bit security in the pairing target field.

BLS12-381. Bowe [6] designed BLS12-381 specifically for zk-SNARKs, achieving two-adicity 32 in both fields through careful parameter selection within the BLS12 family. Our work achieves higher two-adicity (36) using the more flexible Cocks-Pinch method, at the cost of a larger ρ -value (2.0 vs. 1.5).

KSS-18 curves. Kachisa, Schaefer, and Scott [19] construct pairing-friendly curves with embedding degree 18 using the Brezing-Weng method [20], achieving $\rho \approx 1.33$. While KSS-18 offers better bandwidth efficiency than Cocks-Pinch ($\rho = 2$), the parametric family does not inherently guarantee NTT-friendliness, and the fixed polynomial structure limits the ability to optimize for readability across a large candidate space.

Pasta curves. The Pallas/Vesta cycle [21] achieves NTT-friendliness through a 2-cycle of curves, enabling recursive proof composition. Unlike our approach, Pasta curves are not pairing-friendly and target a different application domain (Halo 2 recursive proofs).

Nothing-up-my-sleeve numbers and SafeCurves. The cryptographic community has long advocated for transparent parameter generation to demonstrate absence of backdoors. The SafeCurves project [22] formalizes criteria including rigidity, twist security, and completeness. Our readability scoring takes a complementary approach: rather than deriving primes from a fixed public constant, we select the most inspectable prime from a large space of equally secure candidates, while the Cocks-Pinch seed provides full reproducibility.

Cocks-Pinch extensions. Chen, Hong, and Wang [23] extend the Cocks-Pinch method to embedding degrees 5–8 with optimal Ate pairing computation. Our work focuses on $k = 18$ with the novel addition of NTT constraints and readability scoring.

Pairing implementation for $k = 18$. Efficient pairing computation requires optimized Miller loop and final exponentiation routines. Beuchat et al. [24] and Aranha et al. [25] demonstrate high-speed implementations for BN curves ($k = 12$) using tower extensions and optimal Ate formulas. For $k = 18$, the final exponentiation decomposes over $\mathbb{F}_{p^{18}} = \mathbb{F}_{p^3}[w]/(w^6 - \xi)$ with a cyclotomic squaring shortcut in the hard part. No production-grade implementation exists for $k = 18$ at the 1024-bit level; this remains our primary

future work direction, building on the Vercauteren [10] optimal pairing framework.

Transparent parameter generation. Beyond SafeCurves [22], the broader cryptographic community has pursued *verifiable parameter pipelines* where every constant is derived from a public, auditable process. The Dual_EC_DRBG incident [7] and subsequent NIST curve scrutiny established the principle that “nothing-up-my-sleeve” seeds must be accompanied by a deterministic derivation path. Our approach satisfies this criterion: the single 86-bit seed T deterministically produces all curve parameters via $\Phi_{18}(T)$ and the CM method, and the readability scoring selects among equally secure candidates without introducing hidden degrees of freedom.

NTT-friendly non-pairing curves. Several recent systems optimize field primes purely for NTT performance without requiring pairings. The Pasta curves (Pallas/Vesta) [21] form a 2-cycle enabling recursive proof composition in Halo 2. Plonky3 [26] uses the Goldilocks prime ($p = 2^{64} - 2^{32} + 1$) and BabyBear ($p = 2^{31} - 2^{27} + 1$) for STARK-based proofs, achieving extremely high NTT throughput at the cost of small field size and no pairing support. Our work targets a fundamentally different design point: pairing-friendly curves with NTT compatibility, enabling protocols (BLS signatures, KZG commitments) that require bilinear maps rather than hash-based or FRI-based proof systems.

7. Conclusion

We presented three contributions to pairing-friendly elliptic curve construction:

1. **NTT-friendly Cocks-Pinch.** By constraining the cyclotomic seed $T \equiv 0 \pmod{2^{12}}$, we guarantee two-adicity ≥ 36 in the scalar field deterministically, while an extended lift search achieves the same for the base field. For comparison, BLS12-381 achieves two-adicity 32 within the BLS12 family; our construction obtains 36 via the Cocks-Pinch method at a different tradeoff point ($\rho = 2.0$ vs. 1.5) and a modest increase in generation time (30–60 s vs. 2–10 s).
2. **Readability-aware prime selection.** Our multi-criteria scoring system produces primes that are more amenable to human inspection without any security degradation, addressing the auditability concern in cryptographic parameter generation.
3. **Curve1024.** We instantiate these techniques into a concrete, publicly verifiable curve: $E : y^2 = x^3 + 41$ over a 1024-bit prime field with embedding degree 18, providing 256-bit ECDLP security and NTT support up to length 2^{36} in both fields. The complete parameter set (field primes, generator point, CM data) is published alongside a Rust implementation with constant-time scalar multiplication (Montgomery ladder over projective coordinates) and a test suite verifying correctness and attack resistance.

The system is suitable for applications requiring long-term security guarantees—including interactive signing (14 ms per

signature) and future pairing-based protocols (BLS aggregate signatures, zk-SNARKs) that benefit from the NTT-friendly field structure.

Future work includes: (1) full optimal Ate pairing implementation over $\mathbb{F}_{p^{18}}$, enabling BLS aggregate signatures and zk-SNARK verification directly on Curve1024, and (2) extension to other embedding degrees ($k = 12, 24$) to explore the tradeoff between pairing target size and field element compactness.

A. Availability

The implementation, parameter generation tool, and test suite are publicly available at <https://github.com/Tranduy1dol/curve1024> (tag v1.5.0-alpha) under the MIT license. All curve parameters can be independently reproduced from the seed T using the provided construction example.

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